APPENDIX G: ALGORITHM SELECTION AND COMPARISON

G.1 Algorithm Used in the Project

The implemented predictor uses sliding-window ordinary least-squares linear regression. For each aircraft, the latest stored samples are ordered by time. Separate straight lines are fitted for latitude, longitude and altitude: value(t) = a + bt. The fitted lines are evaluated at the last sample time plus the number of seconds for which the aircraft has been absent from the current feed.

Implemented constraints

- Input window: up to the latest 20 stored trajectory points; at least 4 valid points are required.
- Prediction condition: the aircraft must have been observed in the current session but be absent from the current live/replay snapshot.
- Prediction horizon: 1-180 seconds with the current configuration; older records are not displayed as predictions.
- Fallback: constant speed, heading and vertical rate are used when regression cannot be fitted.
- Safety against feedback: predicted rows are output-only and are not written back into the training tables.

G.2 Why This Method Was Selected

For a local academic tracker, this method provides a useful balance: it is understandable, deterministic, fast enough to run during every API refresh, and able to smooth small variations across several observations. It needs no offline training dataset or GPU. Its main limitation is that a straight-line trend cannot model a sharp turn or a prolonged outage, which is why the prediction age is deliberately capped and the interface visually distinguishes predicted data from live data.

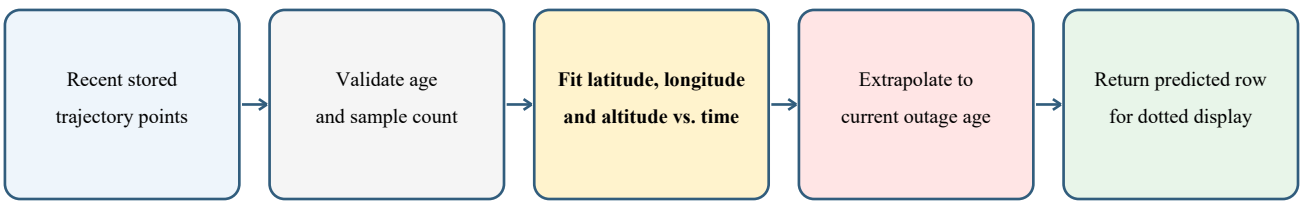


Figure 14: Trajectory Prediction Pipeline